

Lecture 2

EE-215 Electronic Devices and Circuits

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Modeling the Diode Forward Characteristic

- The Exponential Model
- The Constant-Voltage Drop Model
- The Ideal Diode Model
- The Small-Signal Model

The Exponential Model

- provides the most accurate description of diode operation
 - in the forward-bias region
- is the most difficult to use
 - because of its severely non-linear nature
- Exponential model $\Rightarrow i = I_S e^{\frac{v}{V_T}}$

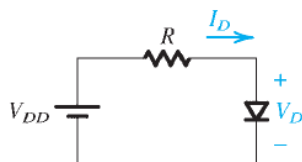


Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

- for this circuit
 - $I_D = I_S e^{\frac{V_D}{V_T}}$
 - and using KVL
 - $V_{DD} = I_D R + V_D$
 - or $I_D = \frac{V_{DD} - V_D}{R}$
 - if I_S for the diode is given,
 - we have two equations in two unknowns I_D and V_D
- $I_D = I_S e^{\frac{V_D}{V_T}}$ and $I_D = \frac{V_{DD} - V_D}{R}$

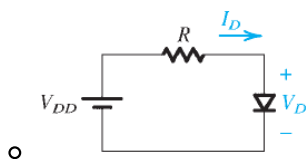


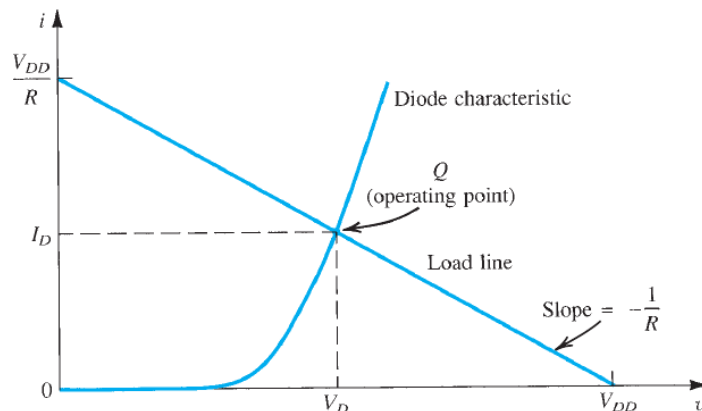
Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

- if I_S for the diode is given,
 - we have two equations in two unknowns I_D and V_D
 - These two equations can be solved using either
 - Graphical Analysis
 - or Iterative Analysis

Graphical Analysis using the Exponential Model

- $I_D = I_S e^{\frac{V_D}{V_T}}$ and $I_D = \frac{V_{DD} - V_D}{R}$
- this analysis is performed by

- plotting the two equations on the i-v plane
- the point of intersection is the desired solution (V_D, I_D)



◦ **Figure 4.11** Graphical analysis of the circuit in Fig. 4.10 using the exponential diode model.

- the curve represents the exponential diode equation.
 - while the straight line represents the KVL equation
 - Such a straight line is called load line.
- The load line intersects the diode curve at point Q
 - Q represents the operating point of the circuit.
 - The coordinates of Q give the values of I_D and V_D

Example 4.4 using Graphical Analysis

- Determine the current I_D and the diode voltage V_D for the circuit in Fig. 4.10 with $V_{DD} = 5V$ and $R = 1k\Omega$. Assume that the diode has a current of $1mA$ at a voltage of $0.7V$.

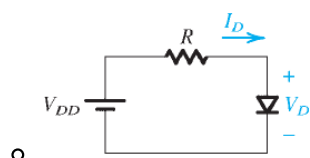
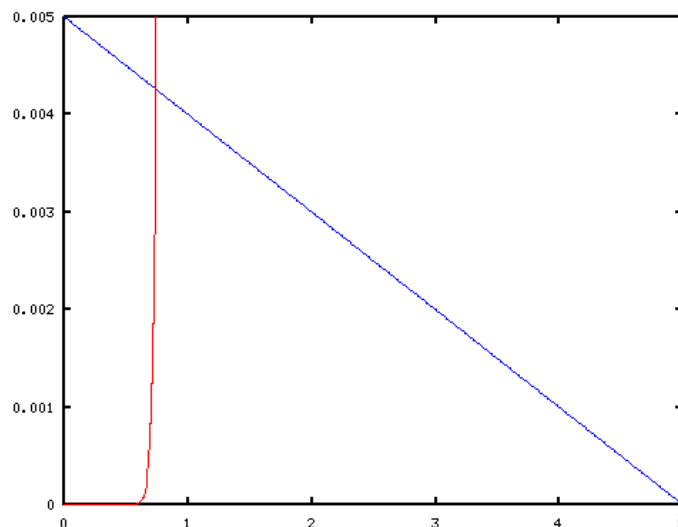


Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

Solution:

- here $V_{DD} = 5V$, $R = 1k\Omega$
 - and $I = 1mA$ for $V = 0.7V$
 - $I = I_S e^{\frac{V}{V_T}} \Rightarrow I_S = \frac{I}{e^{\frac{V}{V_T}}} = \frac{1mA}{e^{\frac{0.7V}{25mV}}} = 6.9144 \times 10^{-16}A$
 - As the 2 circuit equations are $I_D = I_S e^{\frac{V_D}{V_T}}$ and $I_D = \frac{V_{DD} - V_D}{R}$
 - $\Rightarrow I_D = 6.9144 \times 10^{-16} e^{\frac{V_D}{25m}}$ and $I_D = \frac{5 - V_D}{1k}$
- $I_D = 6.9144 \times 10^{-16} e^{\frac{V_D}{25m}}$ and $I_D = \frac{5 - V_D}{1k}$
 - Plotting these 2 equations in Matlab or Octave will give the desired solution V_D, I_D



-
- reading V_D, I_D from the plot $\Rightarrow V_D = 0.7358V$
 - and $I_D = 4.264mA$

Iterative Analysis using the Exponential Model

- the 2 circuit equations
 - $I_D = I_S e^{\frac{V_D}{V_T}}$ and $I_D = \frac{V_{DD} - V_D}{R}$
 - can also be solved using a simple iterative procedure

Modeling the Diode Forward Characteristic

- $I_D = I_S e^{\frac{V_D}{V_T}}$ and $I_D = \frac{V_{DD} - V_D}{R}$

Example 4.4 using Iterative Analysis

- Determine the current I_D and the diode voltage V_D for the circuit in Fig. 4.10 with $V_{DD} = 5V$ and $R = 1k\Omega$. Assume that the diode has a current of $1mA$ at a voltage of $0.7V$.

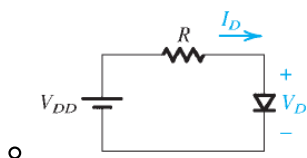


Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

Solution:

- to start, let $V_D = 0.7V$

Iteration 1:

- KVL $\Rightarrow I_D = \frac{V_{DD} - V_D}{R} = \frac{5 - 0.7}{1k} = 4.3mA$
 - to have a better estimate of V_D
 - using $V_2 - V_1 = V_T \ln \frac{I_2}{I_1}$ or $V_2 = V_1 + V_T \ln \frac{I_2}{I_1}$
 - for $V_1 = 0.7V$ and $I_1 = 1mA$ (given)
 - and $V_2 = V_D = ?, I_2 = I_D = 4.3mA$
 - $V_2 = V_1 + V_T \ln \frac{I_2}{I_1} \Rightarrow V_D = 0.7 + 25m \ln \frac{4.3m}{1m} = 0.736V$
 - thus after this first iteration we have
 - $V_D = 0.736V, I_D = 4.3mA$

- Iteration1 $\Rightarrow V_D = 0.736V$, $I_D = 4.3mA$

Iteration 2:

- for the 2nd iteration
 - $I_D = \frac{V_{DD} - V_D}{R} = \frac{5 - 0.736}{1k} = 4.264mA$
 - $V_2 = V_1 + V_T \ln \frac{I_2}{I_1}$
 - here $V_1 = V_D$ from first iteration, $I_1 = I_D$ from first iteration
 - $\Rightarrow V_D = 0.736 + 25m \ln \frac{4.264m}{4.3m} = 0.7358V \approx 0.736V$
 - so 2nd iteration $\Rightarrow V_D = 0.736V$ and $I_D = 4.264mA$
 - As the values obtained in the 2nd iteration
 - are very close to those obtained in the first iteration,
 - further iterations are not required
 - so the solution is $V_D = 0.736V$, $I_D = 4.264mA$

Example 4.4 using Graphical and Iterative Analysis

- Iterative Analysis solution : $V_D = 0.736V$, $I_D = 4.264mA$
 - Note that this solution closely match (exactly match to the 4 decimal),
 - to the one obtained using Graphical Analysis
 - i.e. $V_D = 0.7358V$, $I_D = 4.264mA$ (Graphical Analysis solution)

The Constant Voltage Drop Model

- is the simplest
 - and most widely used model
 - is based on the observation that a conducting diode has
 - voltage drop in a narrow range of 0.6V - 0.8V
 - thus the model assumes a constant value of 0.7V

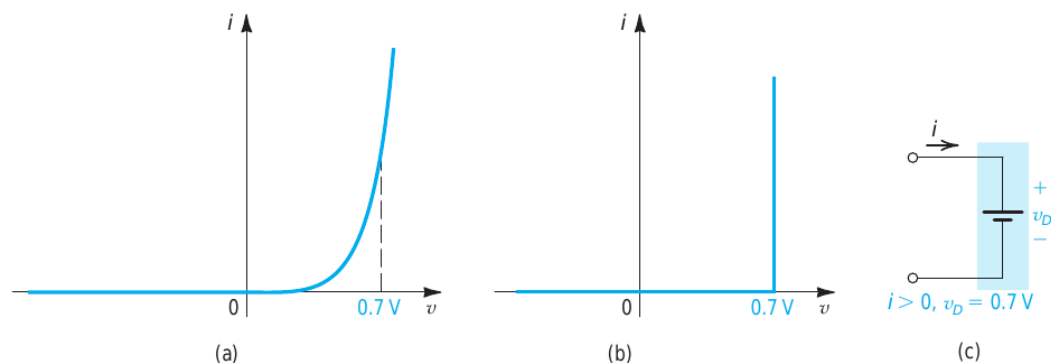


Figure 4.12 Development of the diode constant-voltage-drop model: (a) the exponential characteristic; (b) approximating the exponential characteristic by a constant voltage, usually about 0.7 V; (c) the resulting model of the forward-conducting diodes.

Example 4.4 using The Constant-Voltage-Drop Model

- Determine the current I_D and the diode voltage V_D for the circuit in Fig. 4.10 with $V_{DD} = 5V$ and $R = 1k\Omega$. Assume that the diode has a current of $1mA$ at a voltage of $0.7V$.

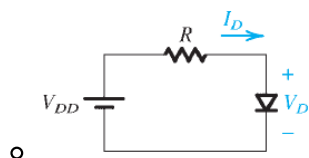


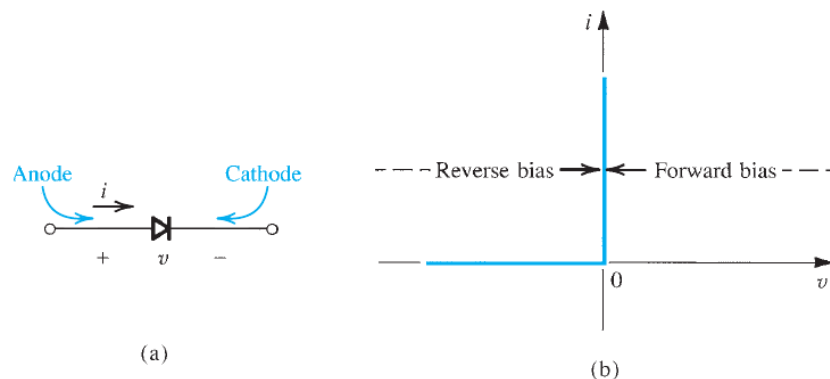
Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

Solution:

- constant voltage drop model $\Rightarrow V_D = 0.7V$
 - KVL $\Rightarrow I_D = \frac{V_{DD} - V_D}{R} = \frac{5 - 0.7}{1k} = 4.3mA$
 - Graphical Analysis solution $\Rightarrow V_D = 0.7358V, I_D = 4.264mA$

The Ideal-Diode Model

- Already studied in detail



- **Figure 4.1** The ideal diode: (a) diode circuit symbol; (b) $i-v$ characteristic;
- The most commonly used model
 - is the constant-voltage-drop model
- The ideal diode model is very useful in determining
 - which diodes are on and which are off
 - in a multi-diode circuit

Example 4.4 using The Ideal-diode Model

- Determine the current I_D and the diode voltage V_D for the circuit in Fig. 4.10 with $V_{DD} = 5V$ and $R = 1k\Omega$. Assume that the diode has a current of $1mA$ at a voltage of $0.7V$.

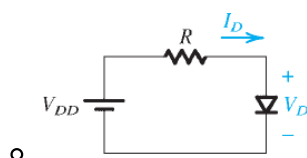


Figure 4.10 A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

Solution:

- ideal diode model $\Rightarrow V_D = 0V$
 - KVL $\Rightarrow I_D = \frac{V_{DD} - V_D}{R} = \frac{5 - 0}{1k} = 5mA$

Modeling the Diode Forward Characteristic

Example 4.4

- Exponential Model using Graphical Analysis $\Rightarrow V_D = 0.7358V, I_D = 4.264mA$
- Exponential Model using Iterative Analysis $\Rightarrow V_D = 0.736V, I_D = 4.264mA$
- Constant-Voltage-Drop model $\Rightarrow V_D = 0.7V, I_D = 4.3mA$
- Ideal-Diode-Model $\Rightarrow V_D = 0V, I_D = 5mA$

Exercise D4.11

- Design the circuit in Fig E4.11 to provide an output voltage of $2.4V$. Assume that the diodes available have $0.7V$ drop at $1mA$.

Solution:

- here $V_1 = 0.7V$, $I_1 = 1mA$
 - required is $V_O = 2.4V$
 - $\Rightarrow V_D = \frac{V_O}{3} = \frac{2.4}{3} = 0.8V$
 - $I_D = ?$
 - $V_2 - V_1 = V_T \ln \frac{I_2}{I_1}$
 - here $V_2 = 0.8V$ and $I_D = I_2 = ?$
 - $V_2 - V_1 = V_T \ln \frac{I_2}{I_1} \Rightarrow 0.8 - 0.7 = V_T \ln \frac{I_D}{1m}$
 - or $0.1 = V_T \ln \frac{I_D}{1m} \Rightarrow \frac{0.1}{25m} = \ln \frac{I_D}{1m}$
 - or $\frac{I_D}{1m} = e^{\frac{0.1}{25m}} \Rightarrow I_D = 1me^{\frac{0.1}{25m}} = 54.6mA$

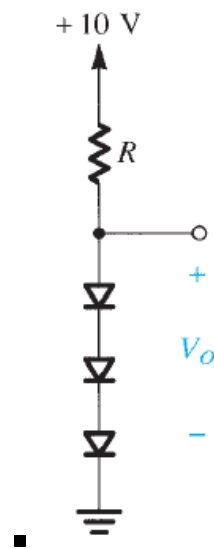


Figure E4.11

- $V_O = 2.4V$
 - $I_D = 54.6mA$
 - As I_D is the current flowing through resistor R
 - Applying ohm's law at R
 - $\Rightarrow I_D = \frac{10-2.4}{R} = 54.6mA$
 - or $R = \frac{10-2.4}{54.6m} = \frac{7.6}{54.6m} = 139.19\Omega$

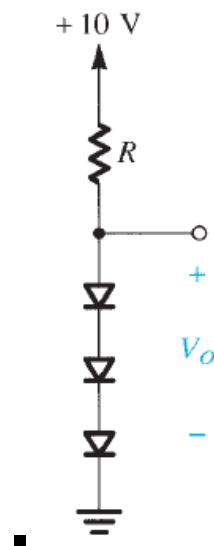
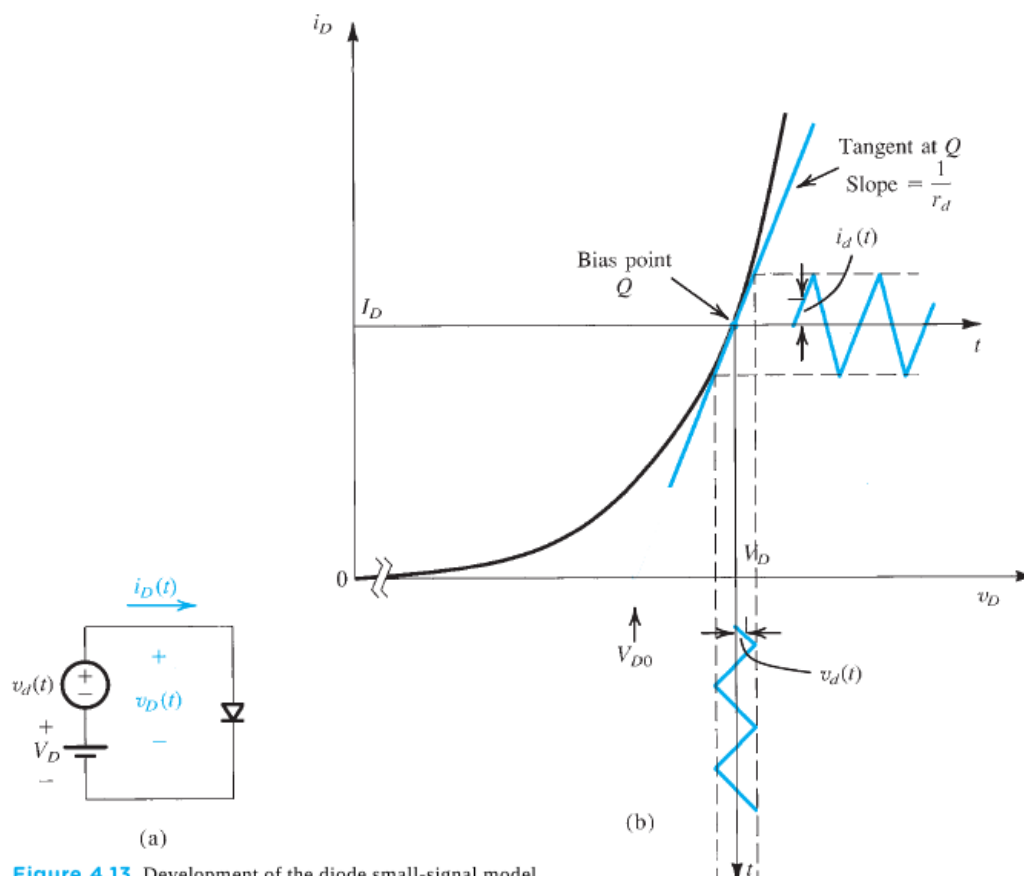


Figure E4.11

The Small-Signal Model

- this model is useful in applications where
 - a diode is forward biased
 - and a small AC signal is superimposed on the DC quantities
 - in such applications,
 - one can first determine the DC operating point (V_D , I_D) of the diode
 - then for small signal operation, around the DC bias point,
 - the diode is modeled by a resistance equal to the inverse of the slope of the tangent to the exponential i-v characteristic at the bias point.



- **Figure 4.13** Development of the diode small-signal model.
- here a dc voltage V_D , represented by a battery is applied to the diode
 - and a time-varying signal $v_d(t)$, is superimposed on the dc voltage V_D
 - In the absence of $v_d(t)$, the diode voltage equals V_D
 - and the diode current is $I_D = I_S e^{V_D/V_T}$

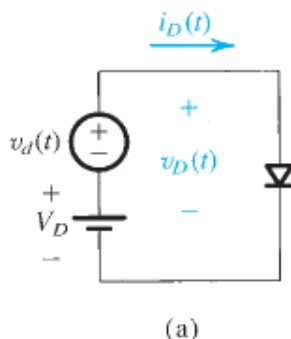
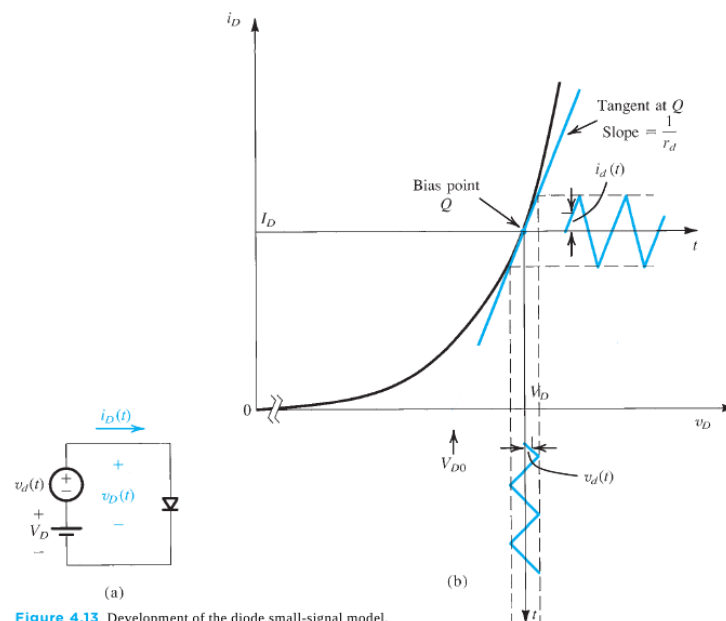


Figure 4.13 Development of the diode small-signal model.

- when the signal $v_d(t)$ is applied, the total instantaneous diode voltage $v_D(t)$ is
 - $v_D(t) = V_D + v_d(t)$
- and the total current is $\Rightarrow i_D(t) = I_S e^{v_D(t)/V_T}$
- $i_D(t) = I_S e^{v_D(t)/V_T}$
 - $i_D(t) = I_S e^{(V_D + v_d)/V_T} \because v_D(t) = V_D + v_d(t)$
 - $i_D(t) = I_S e^{V_D/V_T} e^{v_d/V_T}$
 - $i_D(t) = I_D e^{v_d/V_T} \because I_D = I_S e^{V_D/V_T}$
- As in terms of series, exponential can be represented as
 - $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$
 - $\Rightarrow i_D(t) = I_D e^{v_d/V_T} = I_D \left\{ 1 + \frac{v_d}{V_T} + \frac{1}{2!} \left(\frac{v_d}{V_T} \right)^2 + \dots \right\}$
- $i_D(t) = I_D \left\{ 1 + \frac{v_d}{V_T} + \frac{1}{2!} \left(\frac{v_d}{V_T} \right)^2 + \dots \right\}$
 - now if the amplitude of the signal $v_d(t)$ is kept sufficiently small i.e. $\frac{v_d}{V_T} \ll 1$
 - $\Rightarrow \left(\frac{v_d}{V_T} \right)^2 \ll \frac{v_d}{V_T}$
 - $\Rightarrow \left\{ 1 + \frac{v_d}{V_T} + \frac{1}{2!} \left(\frac{v_d}{V_T} \right)^2 + \dots \right\} \approx \left\{ 1 + \frac{v_d}{V_T} \right\}$
 - $\Rightarrow i_D(t) = I_D \left\{ 1 + \frac{v_d}{V_T} + \frac{1}{2!} \left(\frac{v_d}{V_T} \right)^2 + \dots \right\} \approx I_D \left\{ 1 + \frac{v_d}{V_T} \right\}$
 - this is the small signal approximation.
 - and this approximation is valid for signals whose amplitudes are smaller than about 5mV
 - e.g. if $v_d = 5mV \Rightarrow \frac{v_d}{V_T} = \frac{5m}{25m} = 0.2$
 - 3rd term in series = $\frac{1}{2!} \left(\frac{v_d}{V_T} \right)^2 = \frac{1}{2!} (0.2)^2 = 0.02$ which is 10 times smaller than 2nd term and can be ignored.
- $i_D(t) \approx I_D \left\{ 1 + \frac{v_d}{V_T} \right\}$
 - or $i_D(t) = I_D + \frac{I_D}{V_T} v_d$
 - thus we have a signal (ac) current ($i_d = \frac{I_D}{V_T} v_d$) superimposed on the dc current I_D
 - i.e. $i_D = I_D + i_d$
 - where $i_d = \frac{I_D}{V_T} v_d$
 - thus the signal current i_d is directly proportional to the signal voltage v_d
 - this quantity relating the signal current i_d to the signal voltage v_d is called the small-signal conductance.
 - small-signal conductance = $\frac{i_d}{v_d} = \frac{I_D}{V_T}$
 - the inverse of this conductance is the diode small-signal resistance (also called diode incremental resistance)
- diode small-signal resistance = $r_d = \frac{V_T}{I_D} = \frac{v_d}{i_d}$
 - thus r_d is inversely proportional to the diode bias current I_D
 - One can see the small signal approximation from the graphical representation

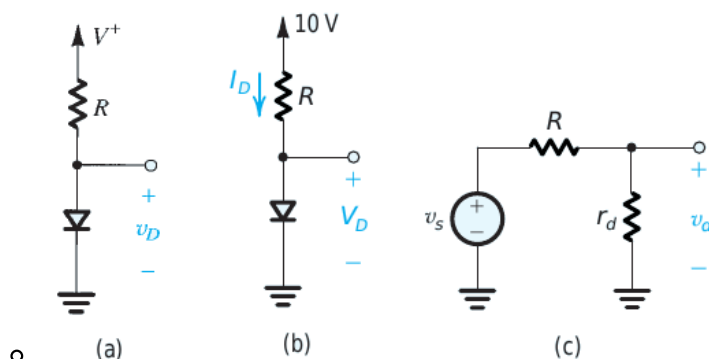


○ Figure 4.13 Development of the diode small-signal model.

- Using the small-signal approximation is equivalent to
 - assuming that the signal amplitude is sufficiently small such that
 - the signal swing along the i-v curve is limited to a short almost-linear segment
 - the slope of this linear segment
 - equals the slope of the tangent to the i-v curve at the operating point Q
 - equals the small signal conductance
 - $\Rightarrow \frac{1}{r_d} = \left[\frac{\partial i_D}{\partial v_D} \right]_{i_D=I_D}$
 - i.e. slope of the i-v curve at $i = I_D$ is equal to $\frac{1}{r_d} = \frac{I_D}{V_T}$
- $\frac{1}{r_d} = \left[\frac{\partial i_D}{\partial v_D} \right]_{i_D=I_D}$
 - $\frac{\partial i_D}{\partial v_D} = \frac{\partial}{\partial v_D} (I_S e^{v_D/V_T}) \because i_D = I_S e^{v_D/V_T}$
 - $\frac{\partial i_D}{\partial v_D} = I_S e^{v_D/V_T} \frac{\partial}{\partial v_D} (v_D/V_T) = I_S e^{v_D/V_T} \left(\frac{1}{V_T} \right)$
 - As $i_D = I_S e^{v_D/V_T}$
 - $\frac{\partial i_D}{\partial v_D} = I_S e^{v_D/V_T} \left(\frac{1}{V_T} \right) = \frac{i_D}{V_T}$
 - $\Rightarrow \frac{1}{r_d} = \left[\frac{\partial i_D}{\partial v_D} \right]_{i_D=I_D} = \left[\frac{i_D}{V_T} \right]_{i_D=I_D} = \frac{I_D}{V_T}$
- The small signal model
 - $\Rightarrow$ dc and ac analysis can be performed independently
 - first dc analysis is carried out, which leads to the Q point
 - then the small signal equivalent circuit is obtained by
 - eliminating all dc sources
 - i.e. replacing dc voltage source by short circuit
 - replacing dc current source by open circuit
 - and replacing the diode by its small-signal resistance $r_d = \frac{V_T}{I_D}$

Example 4.5

- Consider the circuit shown in Fig 4.14(a) for the case in which $R = 10k\Omega$. The power supply V^+ has a dc value of 10 V on which is superimposed a 60-Hz sinusoid of 1-V peak amplitude. (This “signal component” of the power supply voltage is an imperfection in the power-supply design. It is known as the power-supply ripple.) Calculate the dc voltage of the diode and the amplitude of the sine-wave signal appearing across it. Assume the diode to have a 0.7-V drop at 1 mA current.

**Figure 4.14**

(a) Circuit for Example 4.5.

(b) Circuit for calculating the dc operating point.

(c) Small-signal equivalent circuit.

- using the small signal model $\Rightarrow$ dc analysis can be performed first followed by small-signal analysis
- dc analysis

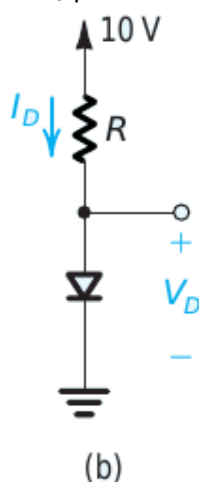
- let $V_D \approx 0.7V$

- ohm's law across $R \Rightarrow$

- $I_D = \frac{10-0.7}{R} = \frac{9.3}{10k} = 0.93mA$

- As this value is very close to 1mA, the diode voltage will be very close to the assumed 0.7V

- $\Rightarrow$ the Q point is $V_D = 0.7V$, $I_D = 0.93mA$

**Figure 4.14**

(b) Circuit for calculating the dc operating point.

- ac analysis

- the Q point is $V_D = 0.7V$, $I_D = 0.93mA$

- At this operating point, r_d is

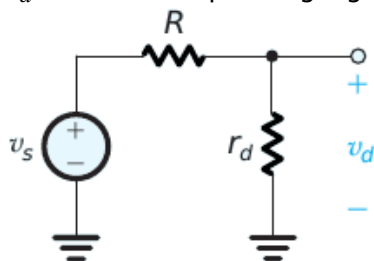
- $r_d = \frac{V_T}{I_D} = \frac{25mV}{0.93mA} = 26.9\Omega$

- small-signal equivalent circuit can be drawn by replacing 10V voltage source by short-circuit and diode by r_d

- in the figure

- v_s is 60Hz 1-V peak sinusoidal component of V^+

- v_d is the corresponding signal across the diode



(c)

Figure 4.14

(c) Small-signal equivalent circuit.

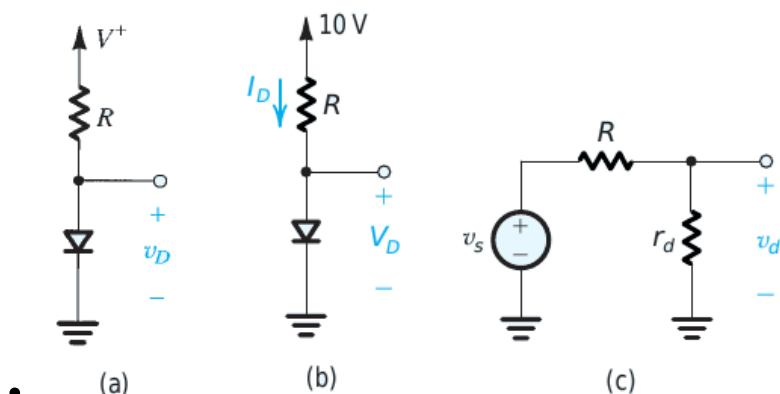


Figure 4.14

(a) Circuit for Example 4.5.

(b) Circuit for calculating the dc operating point.

(c) Small-signal equivalent circuit.

- by voltage divider rule (fig c)
- $v_d = \frac{r_d}{r_d + R} v_s = \frac{26.9}{26.9 + 10k} v_s = (2.683m) v_s$
- for $v_{s,peak} = 1V$

$$\blacksquare v_{d,peak} = 2.683mV$$

- Note that this $v_{d,peak} < 5mV$

■ $\implies$ the small-signal model is valid for this problem

Use of the Diode Forward Drop in Voltage Regulation

A voltage Regulator

- is a circuit which can provide a constant dc voltage between its output terminals
 - inspite of the changes in the load current drawn from the regulator output terminals
 - or change in the dc power supply voltage that feeds the regulator circuit.
 - As the forward-voltage drop of the diode remains
 - almost constant at approximately 0.7V,
 - while the current changes by relatively large amounts,
 - a forward-biased diode can be used as a simple voltage regulator
 - Regulated voltages greater than 0.7V can be obtained by connecting diodes in series.

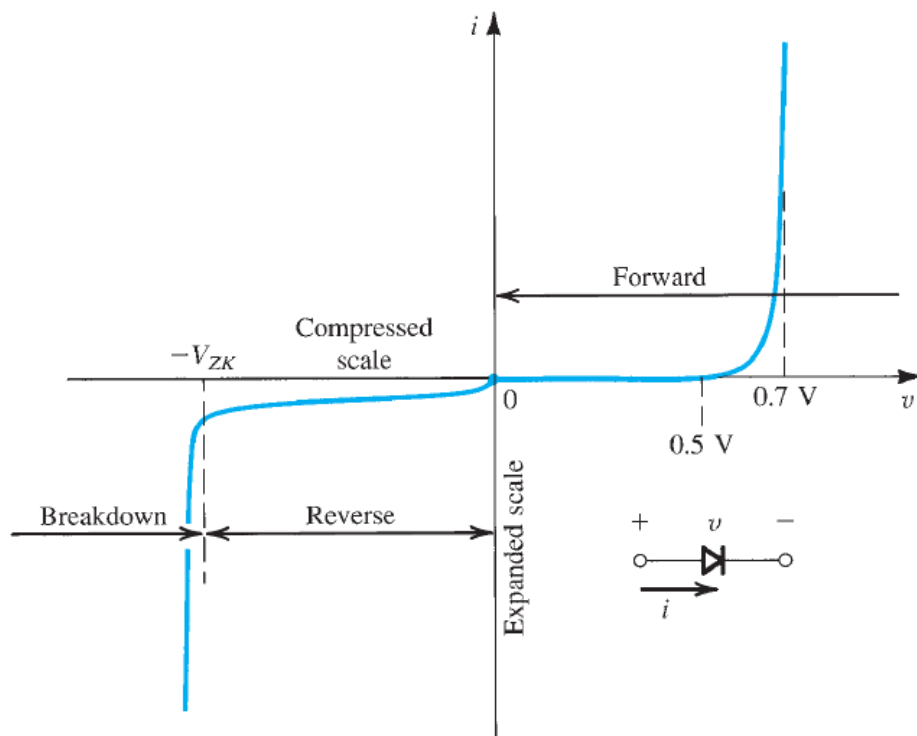


Figure 4.8 The diode i - v relationship with some scales expanded and others compressed in order to reveal details.

Example 4.6

- Consider the circuit shown in Fig. 4.15. A string of three diodes is used to provide a constant voltage of about 2.1 V. We want to calculate the percentage change in this regulated voltage caused by (a) a $\pm 10\%$ change in the power-supply voltage and (b) connection of a $1\text{ k}\Omega$ load resistance.

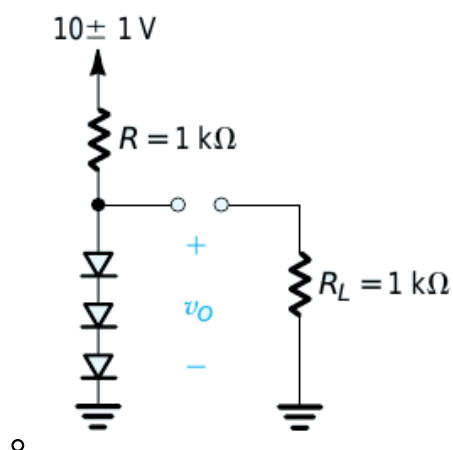
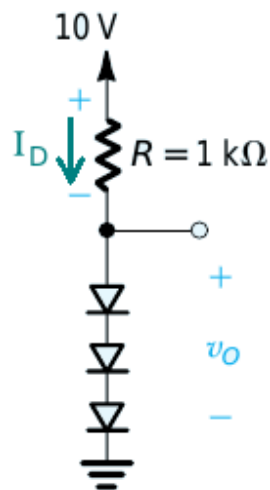


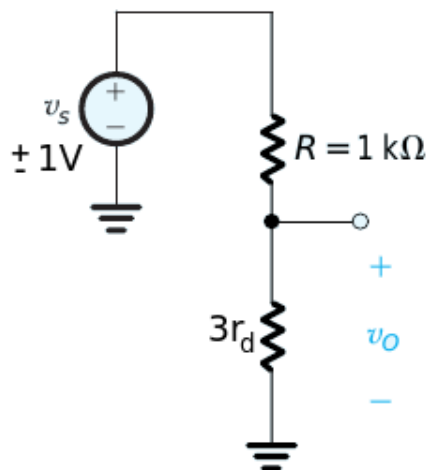
Figure 4.15 Circuit for Example 4.6.

Solution

- no load
 - $V_O = 2.1\text{ V}$
 - $I_R = I_D$
- DC analysis
 - ohm's law at R
 - $\Rightarrow I_D = \frac{10 - V_O}{R} = \frac{10 - 2.1}{1\text{ k}} = 7.9\text{ mA}$
 - thus Q point is $I_D = 7.9\text{ mA}$, $V_D = \frac{V_O}{3} = \frac{2.1}{3} = 0.7\text{ V}$



- Q point is $I_D = 7.9mA$, $V_D = 0.7V$
- AC analysis
 - $r_d = \frac{V_T}{I_D} = \frac{25m}{7.9m} = 3.2\Omega$
 - ac model is obtained by replacing
 - 10V dc source by short circuit
 - and each diode by a resistor of $3.2\Omega (= r_d)$
 - using voltage divider
 - $v_o = \frac{3r_d}{3r_d+1k} v_s = \frac{9.6}{9.6+1k} v_s = (9.5m) v_s$
 - $\Rightarrow v_{o,peak} = (9.5m) v_{s,peak}$
 - for $v_{s,peak} = 1V \Rightarrow v_{o,peak} = 9.5mV$



Example 4.6a

- $v_{s,peak} = 1V \Rightarrow v_{o,peak} = 9.5mV$
 - the percentage change in supply voltage is $= \frac{1V}{10V} \times 100 = 10\%$
 - the corresponding percentage change in output voltage is $= \frac{9.5mV}{2.1V} \times 100 = 0.45\%$
 - which is quite small
 - $\Rightarrow$ the diode is providing voltage regulation.
 - i.e. corresponding to $\pm 1V$ ($\pm 10\%$) change in supply voltage,
 - the output voltage changes only by $\pm 9.5mV$ ($\pm 0.45\%$)
 - Note that

- peak signal voltage across the three diodes is 9.5mV
- $\Rightarrow$ the peak signal voltage across each diode is $\frac{9.5mV}{3} = 3.2mV < 5mV$
- thus the small signal model is valid here

Example 4.6b

- Consider the circuit shown in Fig. 4.15. A string of three diodes is used to provide a constant voltage of about 2.1 V. We want to calculate the percentage change in this regulated voltage caused by (b) connection of a 1 k Ω load resistance.

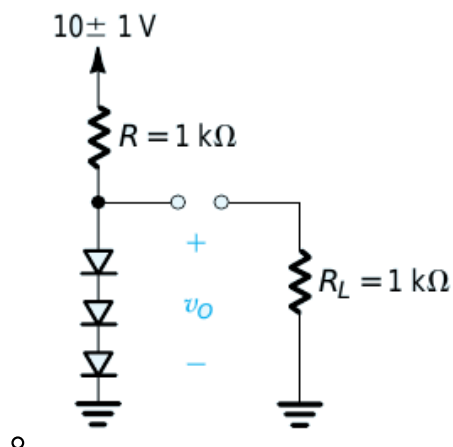
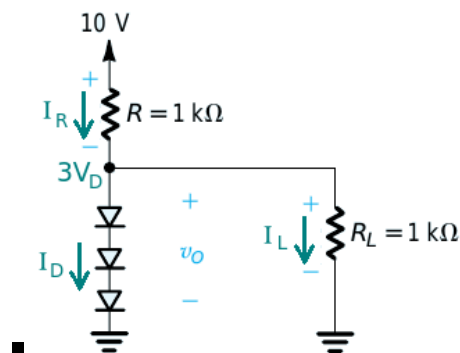


Figure 4.15 Circuit for Example 4.6.

- this part b can be approximately solved by
 - using the small signal model
 - connecting a load resistance of 1k Ω
 - $\Rightarrow$ 1k resistor draws a current of approximately $\frac{2.1}{1k} \approx 2.1mA$
 - $\Rightarrow$ a decrease in diode current by 2.1mA i.e. $\Delta i_d = -2.1mA$
 - $\Rightarrow$ voltage across each diode changes by $\Delta v_d = \Delta i_d r_d = (-2.1m)3.2 = -6.72mV$
 - $\Delta v_o = 3\Delta v_d = -20.16mV$
 - Note that the voltage change across each diode is 6.72mV > 5mV
 - thus using small-signal model in this part b is not entirely justified.

Example 4.6b; continued (Using Exponential Model)

- Accurate solution of this part b can be obtained by
 - using exponential model (graphical analysis or iterative analysis)
 - $I_R = I_D + I_L$
 - $\frac{10-3V_D}{1k} = I_D + \frac{3V_D-0}{1k}$
 - $\frac{10-3V_D}{1k} - \frac{3V_D}{1k} = I_D$
 - $I_D = \frac{10-3V_D-3V_D}{1k} = \frac{10-6V_D}{1k}$
 - and the diode characteristic equation is
 - $V_2 - V_1 = V_T \ln \frac{I_2}{I_1}$



- for iterative analysis

- starting point from part (a) is $V_1 = 0.7V$, $I_1 = 7.9mA$

- let $V_D = 0.7V$

- $I_D = \frac{10-6V_D}{1k}$, $V_2 = V_1 + V_T \ln \frac{I_2}{I_1}$

- for Iteration 1: $V_1 = 0.7V$, $I_1 = 7.9mA$, $V_D = V_1 = 0.7V$

Iteration 1

- $I_D = \frac{10-6V_D}{1k} = \frac{10-6(0.7)}{1k} = 5.8mA$

- $V_2 = V_D = V_1 + V_T \ln \frac{I_2}{I_1} = 0.7 + 25m \ln \frac{5.8m}{7.9m} = 0.69227V$

- for Iteration 2: $V_1 = 0.69227V$, $I_1 = 5.8mA$

- $I_D = \frac{10-6V_D}{1k} = \frac{10-6(0.69227)}{1k} = 5.85mA$

- $V_2 = V_D = V_1 + V_T \ln \frac{I_2}{I_1} = 0.69227 + 25m \ln \frac{5.85m}{5.8m} = 0.69248V$

- Thus after iteration 2, we have $I_D = 5.85mA$, $V_D = 0.69248V$

- $\Rightarrow v_O = 3V_D = 2.0774$

- $\Delta v_O = 2.0774 - 2.1 = -22.6mV$

Operation in the Reverse Breakdown Region - Zener Diodes

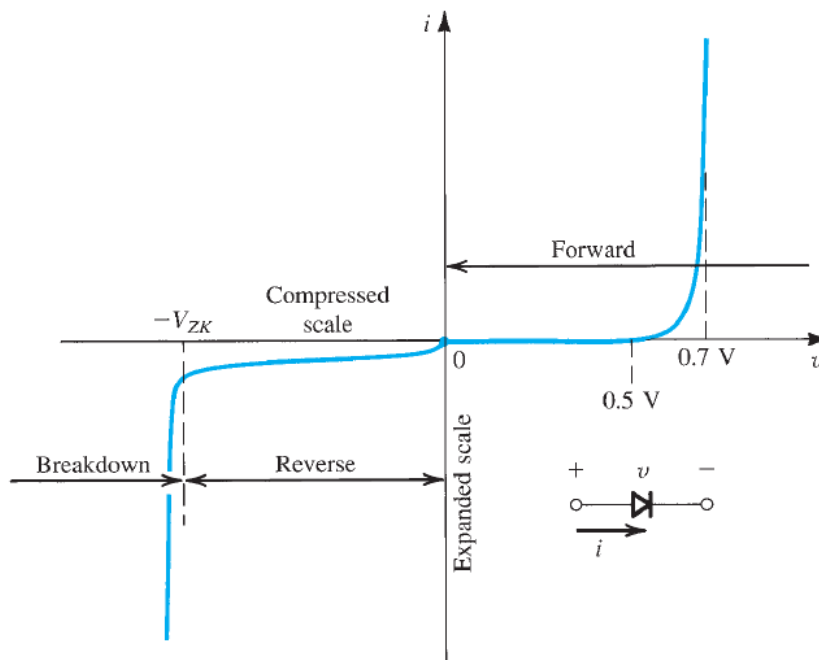


Figure 4.8 The diode i - v relationship with some scales expanded and others compressed in order to

- reveal details.
- In the breakdown region
 - the diode has a very steep i - v curve
 - $\implies$ that it can be used for voltage regulation
- A voltage regulator is a circuit
 - that provides a constant dc output voltage
 - inspite of changes in its load current
 - or changes in its feeding power-supply voltage
- For such voltage regulator applications,
 - special diodes are manufactured to operate specifically in the breakdown region
 - such diodes are called zener diodes (or Breakdown diodes)
 - Zener diodes are fabricated with voltages in the range of a few volts to a few hundred volts.
- the circuit symbol for the zener diodes is

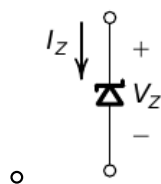


Figure 4.16 Circuit symbol for a zener diode.

- for Zener diodes
 - current flows into the cathode,
 - and the cathode is +ve with respect to the anode.
 - i.e. the zener diode is reverse biased
- thus I_Z and V_Z both have +ve values

Specifying and Modeling the Zener Diode

- for currents greater than the knee current I_{ZK} ,
 - the i - v characteristic is almost a straight line.
 - I_{ZK} is specified on the datasheet of the zener diode.
 - Also the voltage across the zener diode V_Z at a

- specified test current I_{ZT} is given in datasheet.
- from the fig, one can see that
 - corresponding to current change ΔI ,
 - the zener diode voltage changes by ΔV

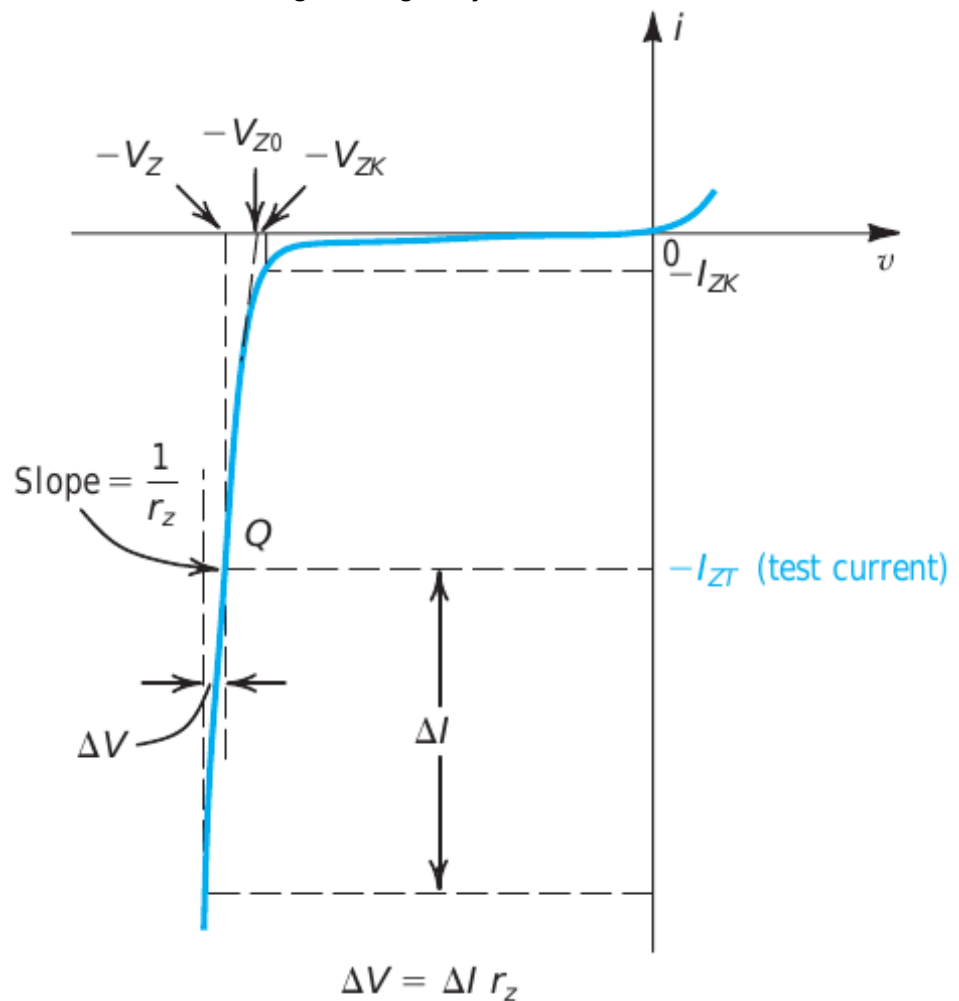


Figure 4.17 The diode i - v characteristic with the breakdown region shown in some detail.

- and this change in voltage can be given as
 - $\Delta V = r_z \Delta I$
- where
 - r_z is the inverse of the slope of the almost linear i - v curve at the operating point Q
 - r_z is called the incremental resistance of the zener diode at point Q
 - r_z is also called dynamic resistance of the zener diode at point Q

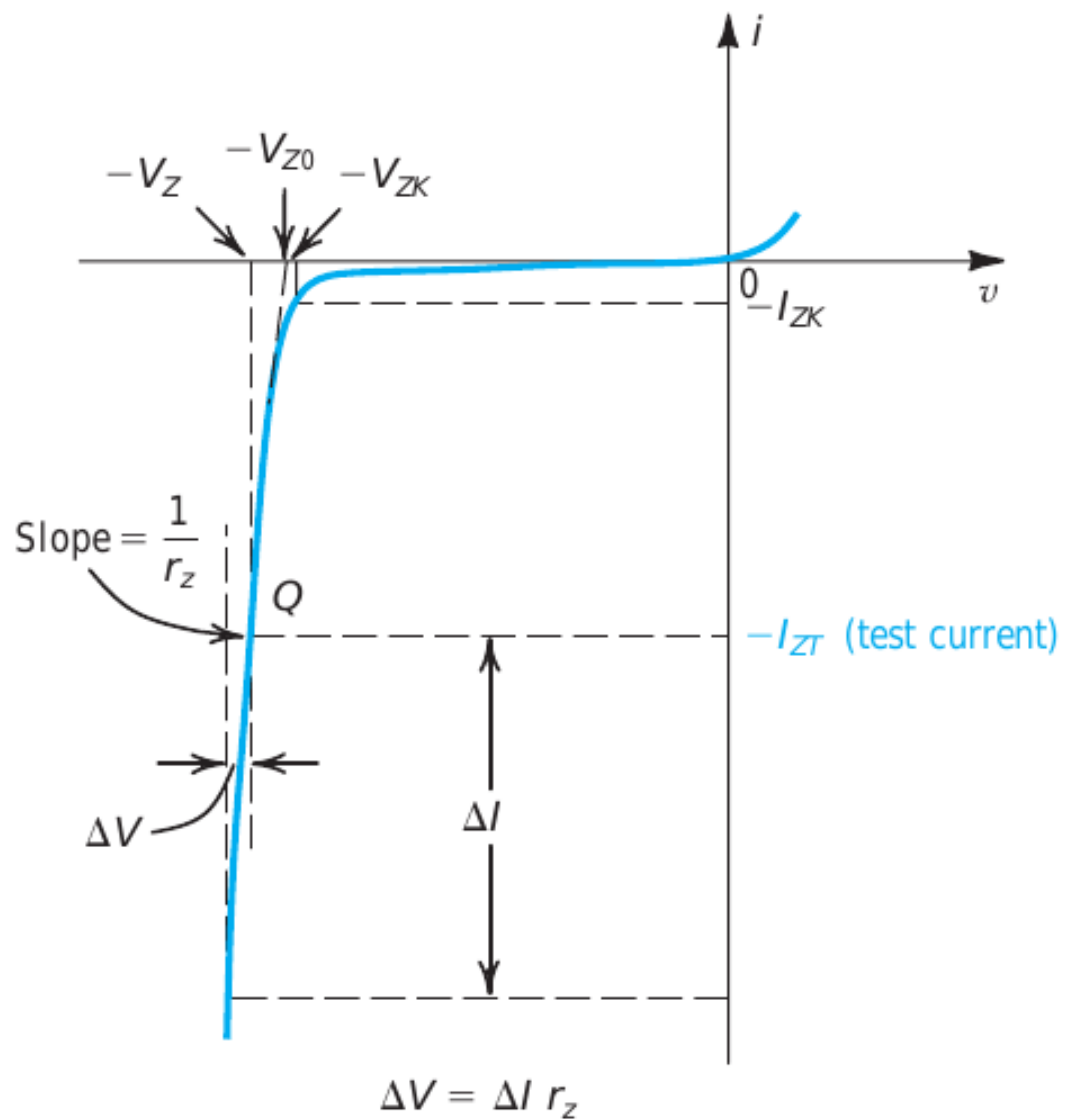


Figure 4.17 The diode i - v characteristic with the breakdown region shown in some detail.

- As a consequence of this almost linear i - v characteristic
 - of the zener diode, it can be modeled as

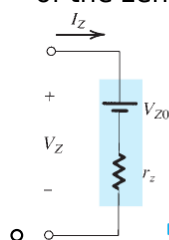


Figure 4.18 Model for the zener diode.

- here V_{Z0} denotes the point at which the straight line
 - of slope $\frac{1}{r_z}$ intersects the voltage axis

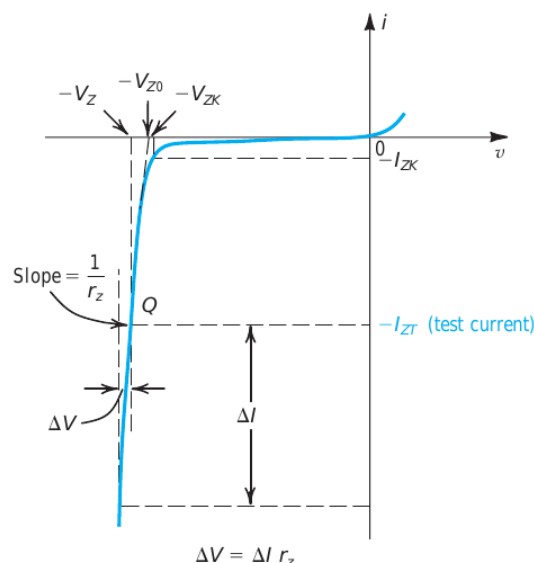


Figure 4.17 The diode i - v characteristic with the breakdown region shown in some detail.

-
- V_{Z0} is only slightly different from the knee voltage V_{ZK}
 - in practice, V_{Z0} and V_{ZK} are almost equal.

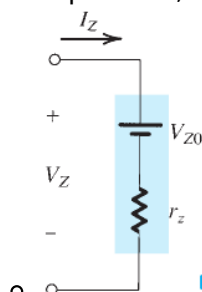


Figure 4.18 Model for the zener diode.

- in equation form, this equivalent circuit can be expressed as
- $V_Z = V_{Z0} + r_Z I_Z$
 - where $I_Z > I_{ZK}$ and $V_Z > V_{Z0}$

Use of the Zener as a Shunt Regulator

- zener diodes can be used for voltage regulation
 - when used for voltage regulation
 - a zener diode appears in parallel (or in shunt) with the load
 - thus such a voltage regulator is called a shunt regulator.

Line Regulation

- Line Regulation = $\frac{\text{change in output voltage}}{\text{change in supply voltage}} = \frac{\Delta V_O}{\Delta V_S}$
 - and is expressed in $\frac{mV}{V}$

Load Regulation

- Load Regulation = $\frac{\text{change in output voltage}}{\text{change in load current}} = \frac{\Delta V_O}{\Delta I_L}$
 - and is expressed in $\frac{mV}{mA}$

Example 4.7

- The 6.8V zener diode in the circuit of Fig 4.19(a) is specified to have $V_Z = 6.8V$ at $I_Z = 5mA$, $r_Z = 20\Omega$, and $I_{ZK} = 0.2mA$. The supply voltage V^+ is nominally 10V but can vary by $\pm 1V$.

- (a) Find V_O with no load and with V^+ at its nominal value.

Solution

- $V^+ = 10V$

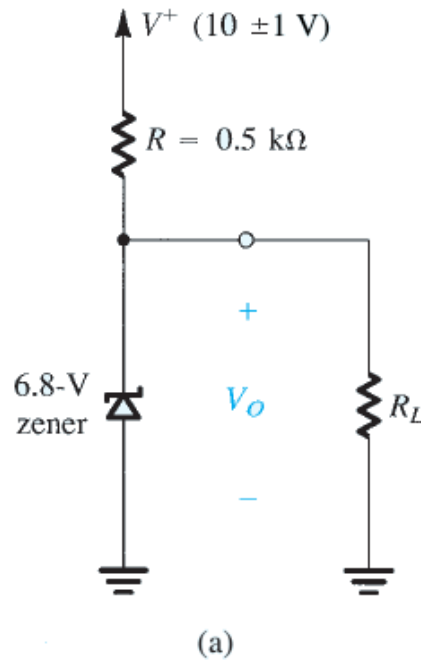


Figure 4.19

- (a) Circuit for Example 4.7.

- $V^+ = 10V$

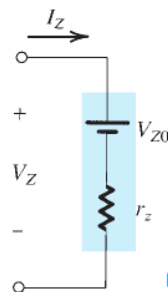


Figure 4.18 Model for the zener diode.

- zener diode model is
- $V_{Z0} = ?$
- As $V_Z = V_{Z0} + r_Z I_Z$
- $\Rightarrow V_{Z0} = V_Z - r_Z I_Z = 6.8 - 20 \times 5m = 6.7V$

- now we can replace the zener diode with its model

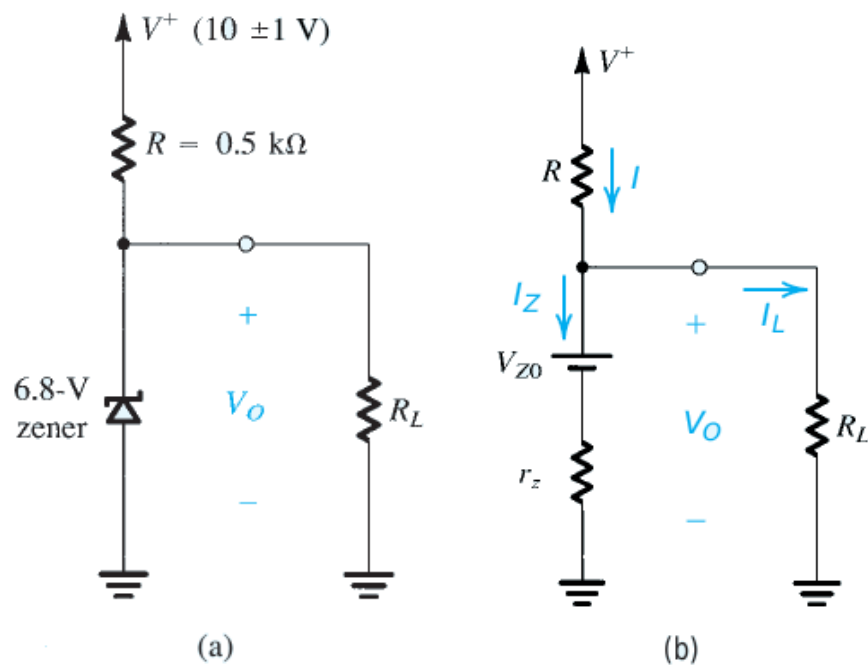


Figure 4.19

◦ (a) Circuit for Example 4.7.

Figure 4.19 (b) The circuit with the zener diode replaced with its equivalent circuit model.

- (a) Find V_O with no load and with V^+ at its nominal value.

Solution 4.7a

- $V^+ = 10V$, $I_L = 0$
 - By KVL
- $V^+ = IR + V_{Z0} + Ir_Z$
 - $I(R + r_Z) = V^+ - V_{Z0}$
 - or $I = I_Z = \frac{V^+ - V_{Z0}}{R + r_Z} = \frac{10 - 6.7}{500 + 20} = 6.35mA$
 - As $V_O = V_Z = V_{Z0} + I_Z r_Z$
 - $V_O = 6.7 + (6.35m)20 = 6.827V$

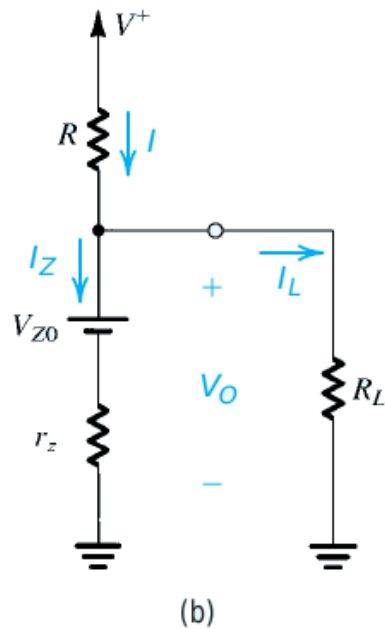
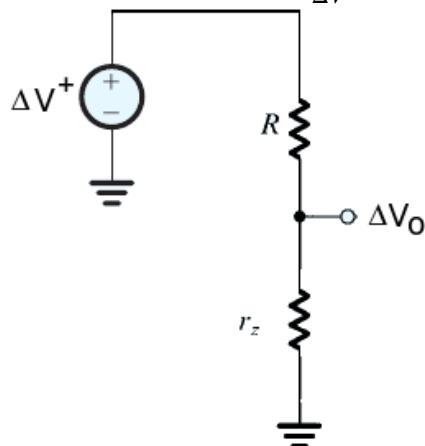


Figure 4.19 (b) The circuit with the zener diode replaced with its equivalent circuit model.

- (b) Find the change in V_O resulting from $\pm 1V$ change in V^+ .

Solution 4.7b

- Line Regulation = $\frac{\Delta V_O}{\Delta V^+} = ?$
- Line Regulation = $\frac{\text{change in output voltage}}{\text{change in supply voltage}}$
 - and is expressed in $\frac{mV}{V}$
 - for a $\pm 1V$ change in V^+ , the corresponding change in output voltage is
 - $\Delta v_O = \frac{r_z}{r_z + R} \Delta V^+ = \frac{20}{20 + 500} (\pm 1V) = \pm 38.5mV$
 - $\Rightarrow$ Line Regulation = $\frac{\Delta V_O}{\Delta V^+} = \frac{38.5mV}{1V} = 38.5 \frac{mV}{V}$

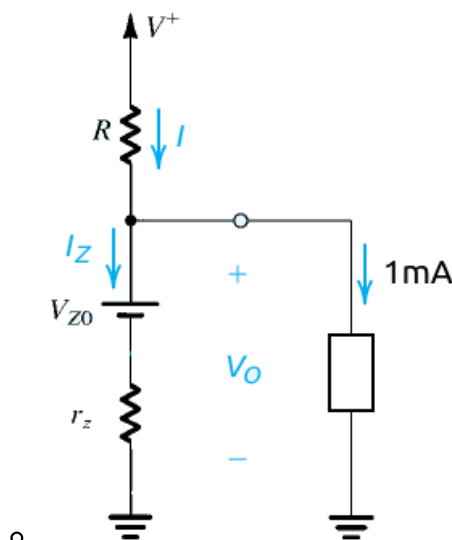


- (c) Find the change in V_O resulting from connecting a load resistance R_L that draws current $I_L = 1mA$.

Solution 4.7c

- KCL $\Rightarrow I = I_z + 1m$
 - $\frac{10 - V_O}{500} = \frac{V_O - V_{z0}}{20} + 1m$

- Multiply 500 on both sides
- $10 - V_o = (V_o - 6.7)25 + 0.5$
- $10 - V_o = 25V_o - 167.5 + 0.5$
- $10 - V_o = 25V_o - 167$
- $10 + 167 = 25V_o + V_o$
- $\frac{177}{26} = V_o$
- $V_o = 6.80769V$
- $V_z = V_o = 6.80769V > V_{zo} \therefore V_{zo} = 6.7V$
- $I_z = \frac{V_o - V_{zo}}{r_z} = \frac{6.80769V - 6.7}{20} = 5.3845mA > I_{zk}$
- As $V_z > V_{zo}$ and $I_z > I_{zk} \implies$ Our assumption is correct i.e. Diode is indeed in Reverse Breakdown region
- Load Regulation = $\frac{\Delta V_o}{\Delta I_L} = ?$
 - Load Regulation = $\frac{\text{change in output voltage}}{\text{change in load current}}$
 - and is expressed in $\frac{mV}{mA}$
 - when a load resistance R_L that draws a load current of $I_L = 1mA$ is connected,
- $\Delta V_o = (V_o \text{ obtained in this part(c)}) - (V_o \text{ from part(a)}) = 6.80769 - 6.927$
- $\Delta V_o = -0.01931V = -19.31mV$
- Load Regulation = $\frac{\Delta V_o}{\Delta I_L} = \frac{-19.31mV}{1mA} = -19.31 \frac{mV}{mA}$

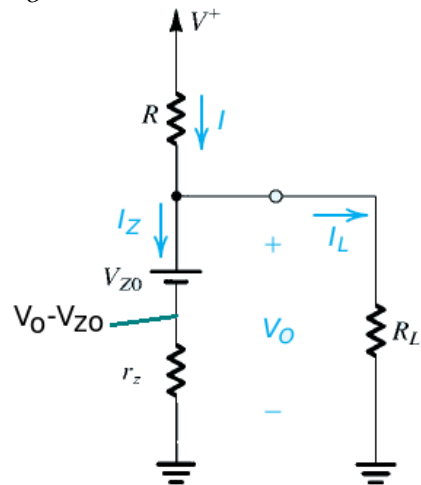


- (d) Find the change in V_o when $R_L = 2k\Omega$.

Solution 4.7d

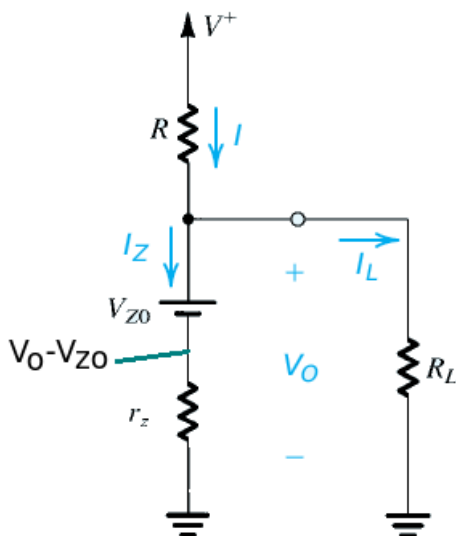
- KCL $\implies I = I_z + I_L$
 - or $\frac{V^+ - V_o}{R} = \frac{V_o - V_{zo}}{r_z} + \frac{V_o}{R_L}$
 - $\frac{V^+}{R} - \frac{V_o}{R} = \frac{V_o}{r_z} - \frac{V_{zo}}{r_z} + \frac{V_o}{R_L}$
- Rearranging
 - $\frac{V^+}{R} + \frac{V_{zo}}{r_z} = \frac{V_o}{R} + \frac{V_o}{r_z} + \frac{V_o}{R_L}$
 - $\frac{V^+}{R} + \frac{V_{zo}}{r_z} = V_o \left(\frac{1}{R} + \frac{1}{r_z} + \frac{1}{R_L} \right)$
 - $V_o = \left(\frac{V^+}{R} + \frac{V_{zo}}{r_z} \right) / \left(\frac{1}{R} + \frac{1}{r_z} + \frac{1}{R_L} \right)$
 - $V_o = \left(\frac{10}{500} + \frac{6.7}{20} \right) / \left(\frac{1}{500} + \frac{1}{20} + \frac{1}{2000} \right)$

◦ $V_O = 6.7619V$



• $V_O = 6.7619V$

- As $V_O > V_{Z0} = 6.7V$, the diode is in breakdown region.
- $\Delta V_O = 6.7619 - 6.83 = -68.095mV$



-
- (e) Find the value of V_O when $R_L = 0.5k\Omega$

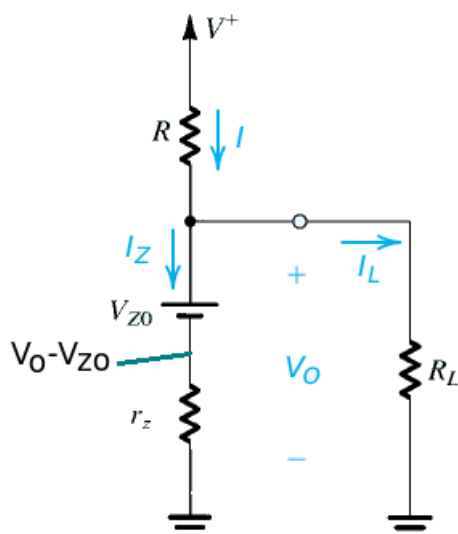
Solution 4.7e

• KCL $\Rightarrow I = I_Z + I_L$

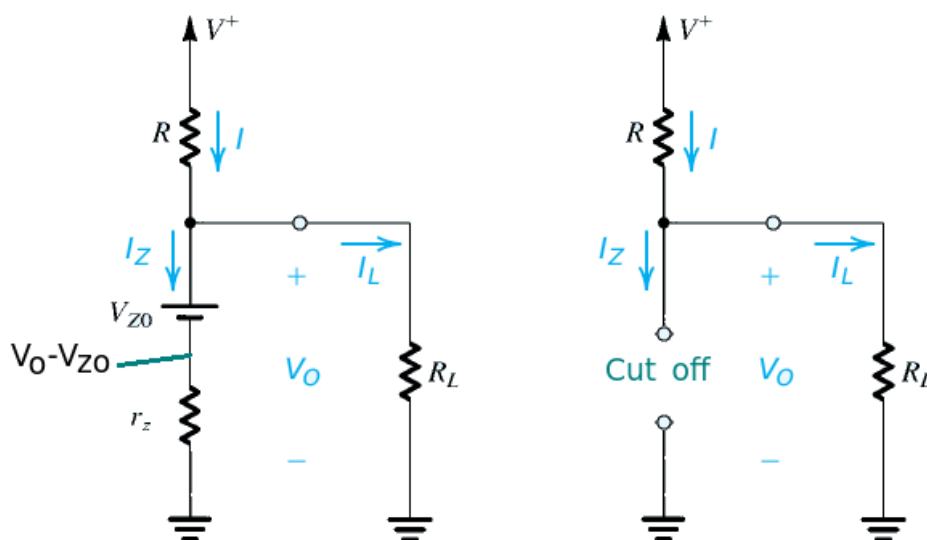
- or $\frac{V^+ - V_O}{R} = \frac{V_O - V_{Z0}}{r_z} + \frac{V_O}{R_L}$
- $\frac{V^+}{R} - \frac{V_O}{R} = \frac{V_O}{r_z} - \frac{V_{Z0}}{r_z} + \frac{V_O}{R_L}$

• Rearranging

- $\frac{V^+}{R} + \frac{V_{Z0}}{r_z} = \frac{V_O}{R} + \frac{V_O}{r_z} + \frac{V_O}{R_L}$
- $\frac{V^+}{R} + \frac{V_{Z0}}{r_z} = V_O \left(\frac{1}{R} + \frac{1}{r_z} + \frac{1}{R_L} \right)$
- $V_O = \left(\frac{V^+}{R} + \frac{V_{Z0}}{r_z} \right) / \left(\frac{1}{R} + \frac{1}{r_z} + \frac{1}{R_L} \right)$



- $V_O = \left(\frac{V^+}{R} + \frac{V_{Z0}}{r_z} \right) / \left(\frac{1}{R} + \frac{1}{r_z} + \frac{1}{R_L} \right)$
 - here $R = 500\Omega$, $r_z = 20\Omega$, $V^+ = 10V$, $V_{Z0} = 6.7V$, $R_L = 500\Omega$
 - $V_O = \left(\frac{10}{500} + \frac{6.7}{20} \right) / \left(\frac{1}{500} + \frac{1}{20} + \frac{1}{500} \right)$
 - $V_O = 6.5741V$
- Note that $V_O = V_Z < V_{Z0}$
 - for a zener diode to operate in the breakdown region $V_Z > V_{Z0}$
 - $\Rightarrow$ the zener diode is cut off (is operating in reverse bias region)
 - $\Rightarrow V_O = \frac{R_L}{R+R_L} V^+ = \frac{500}{500+500} V^+ = \frac{1}{2} V^+$
 - or $V_O = \frac{1}{2} V^+ = 5V$

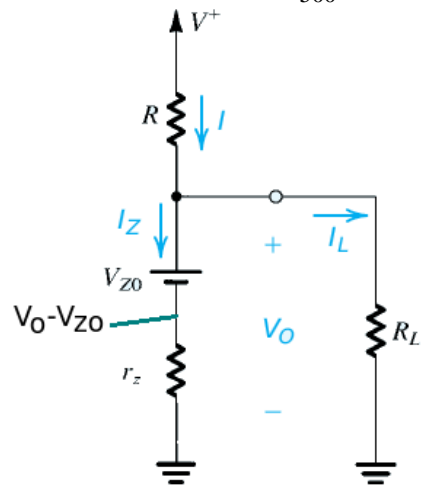


- (f) What is the minimum value of R_L for which the diode still operates in the breakdown region?

Solution 4.7f

- for the zener diode to be in the breakdown region
 - $I_Z > I_{ZK}$, $V_Z > V_{Z0}$
 - thus the lower limit of breakdown region operation is
 - $V_Z = V_{Z0} = 6.7V$
 - and $I_Z = I_{ZK} = 0.2mA$
 - KCL $\Rightarrow I = I_Z + I_L = 0.2mA + I_L$

◦ Ohm's Law $\Rightarrow I = \frac{V^+ - V_O}{500}$



• KCL $\Rightarrow I = I_Z + I_L = 0.2m + I_L$

◦ Ohm's Law $\Rightarrow I = \frac{V^+ - V_O}{500}$

• As $V^+ = 10 \pm 1V$ and its minimum value is 9V

◦ $I = \frac{V^+ - V_O}{500} = 0.2m + I_L$

• $I_L = \frac{V^+ - V_O}{500} - 0.2mA$

◦ $I_L = \frac{9 - 6.7}{500} - 0.2mA = 4.4mA$

◦ $\Rightarrow R_L = \frac{V_L}{I_L} = \frac{6.7}{4.4mA} = 1.52k\Omega$

